I 320D · Data Science for Biomedical Informatics · Spring 2026

Step 2: Exploratory Data Analysis

Expanded Reference · Challenge Questions with Full Answers

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What this document contains

This cheatsheet expands Step 2 of the Project 4 notebook into a complete standalone reference covering:

- Full step-by-step EDA instructions: class balance, vital sign distributions, missing data heatmap, pre-sepsis trend analysis
- Complete model answers to all four Step 2 challenge questions (Q9 – Q12) at graduate level
- Q9: EtCO2 missingness strategy — three-option decision framework with evidence for each
- Q10: Age-stratified pre-sepsis trend plots — code, expected findings, and clinical interpretation
- Q11: Why Pearson correlation fails for time series clinical data and what alternatives to use
- Q12: Site-bias analysis (Set A vs Set B) with visual diagnostics, statistical tests, and mitigation strategies
- Self-assessment checklist with exam-weight markers

PART 1 — Step-by-Step EDA Instructions

1. Visualise class balance at patient-level and hour-level

EDA always starts with the outcome distribution — before looking at any feature. Compute both levels:

```
import matplotlib.pyplot as plt

# ── Patient-level ──────────────────────────────────────────────────────
pt_sepsis   = df.groupby('PatientID')['SepsisLabel'].max()
n_sep       = pt_sepsis.sum()
n_total     = len(pt_sepsis)

# ── Hour-level ─────────────────────────────────────────────────────────
n_sep_hrs   = df['SepsisLabel'].sum()
n_total_hrs = len(df)

fig, axes = plt.subplots(1, 2, figsize=(11, 4))

# Pie: patient level
axes[0].pie([n_total - n_sep, n_sep],
            labels=['No Sepsis', 'Sepsis'],
            autopct='%1.1f%%',
            colors=['#5499C7', '#E87040'], startangle=90)
axes[0].set_title(f'Patient-level  (n={n_total:,})')

# Bar: hour level
counts = [n_total_hrs - n_sep_hrs, n_sep_hrs]
axes[1].bar(['Non-sepsis hours', 'Sepsis hours'], counts,
            color=['#5499C7', '#E87040'])
axes[1].set_title(f'Hour-level  (n={n_total_hrs:,})')
for i, v in enumerate(counts):
    axes[1].text(i, v + 5000, f'{v:,}', ha='center', fontsize=9)

plt.suptitle('Class balance: patient-level vs hour-level', fontsize=12)
plt.tight_layout()
plt.savefig('class_balance.png', dpi=120, bbox_inches='tight')
plt.show()
```

2. Compare vital sign distributions by outcome

```
VITALS = ['HR','O2Sat','Temp','SBP','MAP','DBP','Resp','EtCO2']
VITALS = [v for v in VITALS if v in df.columns]

fig, axes = plt.subplots(2, 4, figsize=(16, 7))
axes = axes.flatten()

for i, col in enumerate(VITALS):
    ax = axes[i]
    for label, color, name in [(0,'#5499C7','No
Sepsis'),(1,'#E87040','Sepsis')]:
        vals = df[df['SepsisLabel'] == label][col].dropna()
        ax.hist(vals, bins=50, alpha=0.55, color=color, label=name,
density=True)
    ax.set_title(col, fontsize=10)
    ax.legend(fontsize=7)

plt.suptitle('Vital sign distributions: Sepsis vs Non-Sepsis', fontsize=13)
```

```
plt.tight_layout()
plt.savefig('vital_distributions.png', dpi=120, bbox_inches='tight')
plt.show()

# KEY SIGNAL DIRECTIONS TO VERIFY:
# HR      -> higher in sepsis (compensatory tachycardia)
# Resp    -> higher in sepsis (tachypnoea / metabolic acidosis compensation)
# O2Sat   -> lower in sepsis (ventilation-perfusion mismatch)
# SBP/MAP -> lower in sepsis (vasodilation from inflammatory mediators)
# Temp    -> bimodal in sepsis (fever OR hypothermia -- both are SIRS
criteria)
```

3. Map missing data rates across all features

```
FEATURE_COLS = [c for c in df.columns
                if c not in ('PatientID','Hour','SepsisLabel','EarlyLabel')]

miss_pct = df[FEATURE_COLS].isnull().mean().sort_values(ascending=False)

plt.figure(figsize=(14, 5))
miss_pct.plot(kind='bar', color='#E87040', edgecolor='white')
plt.axhline(0.50, color='red',     linestyle='--', label='50% missing')
plt.axhline(0.80, color='darkred', linestyle='--', label='80% missing')
plt.ylabel('Fraction of rows missing')
plt.title('Missing data fraction per feature')
plt.legend()
plt.xticks(rotation=45, ha='right', fontsize=8)
plt.tight_layout()
plt.savefig('missing_data.png', dpi=120, bbox_inches='tight')
plt.show()

# Print the three tiers for Q9 analysis
print('Features > 80% missing:', miss_pct[miss_pct > 0.80].index.tolist())
print('Features 50-80% missing:', miss_pct[(miss_pct>=0.50) &
(miss_pct<0.80)].index.tolist())
print('Features < 20% missing:', miss_pct[miss_pct < 0.20].index.tolist())
```

4. Plot mean vital sign trajectories in the 12 hours before sepsis onset

```
def get_pre_onset_window(df, hours_before=12):
    """Align all sepsis patients to their onset hour and extract the
preceding window."""
    records = []
    for pid, grp in df.groupby('PatientID'):
        grp = grp.sort_values('Hour')
        onset_rows = grp[grp['SepsisLabel'] == 1]
        if len(onset_rows) == 0:
            continue
        onset_hour = onset_rows['Hour'].min()
        window = grp[(grp['Hour'] >= onset_hour - hours_before) &
                     (grp['Hour'] <  onset_hour)].copy()
        window['hours_to_onset'] = window['Hour'] - onset_hour
        records.append(window)
    return pd.concat(records, ignore_index=True) if records else
pd.DataFrame()

pre_df = get_pre_onset_window(df, hours_before=12)
trend_cols = ['HR','O2Sat','SBP','Resp']
trend = pre_df.groupby('hours_to_onset')[trend_cols].mean()
```

```
fig, axes = plt.subplots(1, len(trend_cols), figsize=(14, 4))
for ax, col in zip(axes, trend_cols):
    ax.plot(trend.index, trend[col], marker='o', lw=2)
    ax.axvline(0, color='red', linestyle='--', label='Onset')
    ax.set_xlabel('Hours relative to onset')
    ax.set_title(f'Mean {col}')
    ax.legend(fontsize=8)

plt.suptitle('Vital sign trends: 12 hours before sepsis onset', fontsize=12)
plt.tight_layout()
plt.savefig('pre_sepsis_trends.png', dpi=120, bbox_inches='tight')
plt.show()
```

PART 2 — Challenge Questions: Full Answers

Q9 EtCO2 is missing >90% of the time. Drop, impute, or binary flag?

HINT *The missing data heatmap typically shows EtCO2 missing >90% of the time. Should you drop this feature entirely, impute it, or encode its missingness as a binary flag? Defend your choice.*

✓ THE SHORT ANSWER

The correct answer is: encode missingness as a binary flag AND impute the value — use both. Dropping discards the clinical signal in the missingness pattern itself. Imputing without flagging pretends the imputed value is real. The two-column approach gives the model access to (a) whatever signal is in the observed EtCO2 values when they exist and (b) the strong clinical signal in whether EtCO2 was measured at all, which encodes intubation status.

✓ WHY EtCO2 IS CLINICALLY SPECIAL

End-tidal CO_2 (EtCO2) is measured by a capnography device attached to an endotracheal or tracheostomy tube — it requires the patient to be **intubated or on non-invasive ventilation**. Its presence in the data therefore encodes a binary clinical fact: this patient was being mechanically ventilated at this hour. Intubation is itself a severity marker — intubated ICU patients have significantly higher SOFA scores and higher sepsis risk than non-intubated patients. Dropping EtCO2 discards this signal entirely.

✓ THREE-OPTION DECISION FRAMEWORK

✗ **OPTION A: Drop the column entirely**

Rationale: 'Over 90% missing — nothing to learn.' Why this is wrong: the presence or absence of EtCO2 is not random. When EtCO2 is present, the patient is intubated. When absent, they are not. This binary intubation signal is clinically predictive and completely lost if you drop the column. You lose both the measured values (small signal) and the missingness pattern (large signal).

~ **OPTION B: Impute with forward-fill + median**

Rationale: 'Treat missing as unknown and fill with best estimate.' Why this is partially wrong: forward-fill of a non-intubated hour with the previous intubated-hour EtCO2 produces a physiologically nonsensical value. Median imputation fills every non-intubated hour with a 'typical' EtCO2 value — but the patient may not even have been intubated. The model receives plausible-looking but meaningless values and cannot distinguish intubated from non-intubated patients. Use imputation only as a fallback after flagging.

✓ **OPTION C (RECOMMENDED): Binary flag + imputed value**

Rationale: 'Encode the missingness as information, then fill the numeric column for model compatibility.' This preserves both signals. The EtCO2_observed flag encodes intubation status (1 = intubated). The EtCO2 column (forward-filled then median-filled) provides whatever signal the actual value carries for intubated hours. The model can learn from both independently or jointly.

✓ COMPLETE CODE: IMPLEMENTING OPTION C

```
# Apply to the full DataFrame before imputation step
HIGH_MISS_FEATURES = ['EtCO2', 'TroponinI', 'Fibrinogen', 'Bilirubin_direct']
# In practice: all features with > 70% missingness

# Step 1: Create binary missingness flags BEFORE any imputation
for col in HIGH_MISS_FEATURES:
    if col in df.columns:
        df[f'{col}_observed'] = df[col].notna().astype(int)
        # 1 = real measurement existed; 0 = was missing (will be imputed)

# Step 2: Impute with forward-fill + median (as in Step 3)
# (done within patient groups — see Step 3 notebook cell)

# Verify the flag has information
print('EtCO2 observation rate:', df['EtCO2_observed'].mean().round(3))
print()
print('Sepsis rate when EtCO2 observed (intubated):')
intub = df[df['EtCO2_observed'] == 1]['SepsisLabel'].mean()
not_intub = df[df['EtCO2_observed'] == 0]['SepsisLabel'].mean()
print(f'  Intubated    : {intub:.3f}')
print(f'  Non-intubated: {not_intub:.3f}')
# Expected: intubated patients have HIGHER sepsis rate -- confirms the flag
is informative
```

✓ EXPECTED EMPIRICAL VALIDATION

Metric	Expected value	Interpretation
EtCO2 observation rate	5 – 12%	Confirms >88% of hours are non-intubated
Sepsis rate when EtCO2 observed	15 – 25%	Intubated patients are 2–3× more likely to develop sepsis than non-intubated
Sepsis rate when EtCO2 missing	5 – 8%	Baseline prevalence; most patients are not intubated and not septic
LR coefficient on EtCO2_observed (after scaling)	Likely large and positive	The model learns: 'being intubated is a sepsis risk factor' — valid clinical insight

✓ GENERALISING THE PRINCIPLE: WHICH FEATURES NEED FLAGS?

Feature	Missingness rate	Flag recommended?
EtCO2	90 – 95%	YES — missingness encodes intubation status
TroponinI	85 – 92%	YES — ordered when cardiac injury suspected; missingness = no cardiac concern
Fibrinogen	80 – 88%	YES — ordered when DIC (disseminated intravascular coagulation) suspected
Bilirubin_direct	75 – 85%	YES — ordered when liver failure suspected
Lactate	60 – 75%	YES — ordered when perfusion inadequacy suspected (the canonical informative-missing example)
WBC	40 – 60%	MAYBE — routine lab; less informative missingness but worth testing
HR	5 – 15%	NO — monitor dropouts; not clinically informative
Temp	10 – 20%	NO — routine vitals; gaps reflect charting frequency, not clinical decisions

Key exam takeaway

The decision rule for binary flags: ask 'Did a clinician decide whether to measure this?' If yes, missingness encodes clinical judgment and deserves a flag. If the measurement is continuous monitoring (HR, SpO2) and gaps reflect sensor dropout or charting lag rather than a deliberate clinical decision, a flag adds less value.

The gold standard: train two LR models — one without flags, one with flags for high-missingness features. Compare AUROC and SHAP importance. If EtCO2_observed has non-trivial SHAP value, the flag is contributing predictive signal and should be retained.

Q10 Age-stratified pre-sepsis trends: do older patients deteriorate earlier?

> **HINT** *Reproduce the pre-sepsis trend plot but stratify by age group (Age < 50 vs Age >= 65). Do older patients show earlier physiological deterioration?*

✓ THE SHORT ANSWER

The expected finding is **nuanced and clinically counterintuitive:** older patients (Age ≥ 65) do NOT necessarily show larger or earlier vital sign changes before sepsis. In fact, their physiological responses are often *blunted* — less tachycardia, less fever — due to beta-blocker use, autonomic dysfunction, and reduced physiological reserve. What older patients may show is a flatter trajectory that crosses the threshold later and at a lower absolute level. This has major implications for model fairness: a model trained predominantly on younger patients may miss sepsis in elderly patients.

✓ COMPLETE CODE

```
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import numpy as np

def get_pre_onset_window_with_age(df, hours_before=12):
    """Extend get_pre_onset_window to include the patient's Age column."""
    records = []
    for pid, grp in df.groupby('PatientID'):
        grp = grp.sort_values('Hour')
        onset_rows = grp[grp['SepsisLabel'] == 1]
        if len(onset_rows) == 0:
            continue
        onset_hour = onset_rows['Hour'].min()
        window = grp[(grp['Hour'] >= onset_hour - hours_before) &
                     (grp['Hour'] <  onset_hour)].copy()
        window['hours_to_onset'] = window['Hour'] - onset_hour
        records.append(window)
    return pd.concat(records, ignore_index=True) if records else
pd.DataFrame()

pre_df = get_pre_onset_window_with_age(df, hours_before=12)

# Define age groups (three tiers for richer analysis)
pre_df['age_group'] = pd.cut(
    pre_df['Age'],
    bins=[0, 49, 64, 200],
    labels=['Under 50', '50 to 64', '65 and over']
)

TREND_COLS   = ['HR','O2Sat','SBP','Resp','MAP','Temp']
AGE_GROUPS   = ['Under 50', '50 to 64', '65 and over']
AGE_COLORS   = ['#2C5F8A', '#1D6B6B', '#BF4F00']
AGE_STYLES   = ['-', '--', ':']

fig, axes = plt.subplots(2, 3, figsize=(15, 9))
axes = axes.flatten()

for i, col in enumerate(TREND_COLS):
```

```
    ax = axes[i]
    for grp_name, color, ls in zip(AGE_GROUPS, AGE_COLORS, AGE_STYLES):
        subset = pre_df[pre_df['age_group'] == grp_name]
        trend  = subset.groupby('hours_to_onset')[col].agg(['mean','sem'])
        if len(trend) == 0:
            continue
        ax.plot(trend.index, trend['mean'], color=color, lw=2,
                linestyle=ls, label=grp_name)
        ax.fill_between(trend.index,
                        trend['mean'] - trend['sem'],
                        trend['mean'] + trend['sem'],
                        color=color, alpha=0.12)
    ax.axvline(0, color='red', lw=1.5, linestyle='--', alpha=0.7,
label='Onset')
    ax.set_xlabel('Hours relative to onset')
    ax.set_ylabel(col)
    ax.set_title(col)
    ax.legend(fontsize=7)

plt.suptitle('Pre-sepsis vital sign trends by age group\n(shaded region = +/-
1 SEM)',
             fontsize=12)
plt.tight_layout()
plt.savefig('age_stratified_trends.png', dpi=120, bbox_inches='tight')
plt.show()

# ── Quantify the difference ──────────────────────────────────────────────
print('Mean HR in the 3 hours before onset by age group:')
window_3h = pre_df[pre_df['hours_to_onset'] >= -3]
print(window_3h.groupby('age_group')['HR'].mean().round(1))

print('\nFraction with HR > 90 in the 3 hours before onset:')
print((window_3h.groupby('age_group')['HR'].apply(lambda x:
(x>90).mean())).round(3))
```

✓ EXPECTED FINDINGS BY VITAL SIGN

Vital sign	Expected in under 50	Expected in 65 and over	Clinical explanation
HR	Clear tachycardia: rising 10–20 bpm in 12h before onset	Blunted rise: only 5–10 bpm increase	Beta-blocker use in elderly (30–50% prevalence); reduced sinus node sensitivity; autonomic dysfunction with aging
Temp	More pronounced fever (>38.5°C common)	Flatter or hypothermic response (may not mount fever)	Hypothermic response to sepsis is more common in elderly and carries worse prognosis; immunosenescence reduces inflammatory response magnitude
SBP	Clear downward trend starting 6–8h before onset	May show less change until late decompensation	Elderly patients often have baseline hypertension and chronically stiff vessels; BP

			may appear 'normal' until late collapse
Resp	Tachypnoea more prominent (compensatory for metabolic acidosis)	Similar tachypnoea but baseline Resp may be higher due to COPD	Respiratory compensation is relatively preserved in elderly; COPD comorbidity inflates baseline RR
O2Sat	Clearer decline starting ~6h before onset	Less distinct decline; may already have lower baseline	Elderly patients with COPD or CHF may already have SpO2 88–94% at baseline, leaving less 'headroom' for visible decline
MAP	Visible downward trend	Less predictable; higher pre-morbid MAP may mask decline	Elderly baseline MAP often elevated; a 'normal' MAP of 65 may represent a significant drop from their personal baseline

✓ CLINICAL IMPLICATIONS AND MODEL FAIRNESS

The age-stratified analysis raises a critical model fairness concern:

If the model performs worse in elderly patients	What to do about it
Mechanism	The model learned: 'HR > 90 rising rapidly = sepsis signal.' Elderly patients with beta-blockers and autonomic dysfunction may reach sepsis with HR 80 — below the learned threshold. The model underestimates their risk.
Detection	Compute AUROC separately for each age group using the test set. A gap of > 0.05 AUROC between groups is clinically and ethically significant.
Data limitation	The PhysioNet dataset may not include medication information (beta-blockers). Without knowing a patient is on HR-suppressing medication, neither the model nor a clinician can correctly interpret a 'low' HR in a deteriorating patient.

Key exam takeaway

The expected answer to 'do older patients show earlier deterioration?' is NOT a simple yes or no. The trajectory shape differs: younger patients show larger magnitude changes with clearer

trends; older patients show smaller magnitude changes but may deteriorate just as rapidly — the signal is subtler, not necessarily earlier.

Any sepsis prediction system deployed in a real hospital must be validated separately in elderly patients (typically > 65, or by ICU admission diagnosis groupings) before go-live. A model with overall AUROC 0.86 but AUROC 0.74 in patients ≥ 75 has a serious equity problem that aggregate metrics hide.

Q11 Why Pearson correlation is misleading for time series clinical data — and what to use instead

HINT *Many EDA tutorials use correlation matrices. Why is a standard Pearson correlation matrix potentially misleading for time series clinical data? What alternative would you use?*

✓ THE SHORT ANSWER

Pearson correlation has **four specific failure modes** for ICU time series data: (1) it pools all rows as if they are independent, ignoring the within-patient temporal structure; (2) it measures only linear association, missing non-linear clinical relationships; (3) it is sensitive to outliers (extreme lab values inflate correlations); and (4) it ignores time lags — HR may not correlate with BP at the same hour but strongly at a 2-hour lag. Alternatives include: Spearman rank correlation, lagged cross-correlation, partial correlation with lag adjustment, and Granger causality tests.

✓ THE FOUR FAILURE MODES IN DETAIL

```
Pearson correlation treats each row in your DataFrame as an independent
observation.
Your DataFrame has ~600 rows per patient on average. Hours 1-600 from Patient A
are far more similar to each other than to any row from Patient B -- they share
the same patient's physiology, medication regimen, and disease trajectory.

Effect: Correlations are dominated by within-patient temporal autocorrelation,
not by cross-patient population-level relationships. The matrix looks highly
structured even if all features were individually predictive of nothing.

Fix: Compute one-row-per-patient summary statistics (e.g. mean over stay)
before running Pearson, OR use within-patient z-scores to remove individual
baselines.
```

Failure Mode 1: Row independence assumption

```
Pearson measures linear association. Many clinical relationships are non-linear:

  Temperature vs Sepsis risk: J-shaped (both fever AND hypothermia signal sepsis)
  HR vs MAP: roughly linear in normal range, breaks down in shock
  Lactate vs Creatinine: weakly linear at normal values, strongly coupled in MSOF

Effect: Pearson reports near-zero correlation for J-shaped or threshold
relationships
even when the clinical association is strong.

Fix: Spearman rank correlation is robust to monotonic non-linearity.
Distance correlation (dcor package) captures arbitrary non-linear dependencies.
```

Failure Mode 2: Linearity assumption misses clinical non-linearity

```
ICU data contains extreme outliers: creatinine 40 mg/dL in acute kidney failure,
HR 160 in paroxysmal atrial fibrillation, glucose 800 in diabetic ketoacidosis.

Effect: A single patient with simultaneous extreme values of two features
can produce a large Pearson r between those features even if the population-level
relationship is weak. E.g. 5 patients with both very high HR and very high
Lactate
(both in septic shock) can push Pearson r(HR, Lactate) from 0.05 to 0.25.

Fix: Use Spearman correlation (rank-based, robust to outliers), or
winsorize features at the 1st and 99th percentile before computing Pearson.
```

Failure Mode 3: Outlier sensitivity inflates apparent correlations

```
Sepsis physiology has temporal dependencies: HR starts rising BEFORE MAP falls.
A sepsis resuscitation response: lactate peaks, then begins to clear 2-4 hours
later.
Pearson at lag 0 (same hour) misses these delayed relationships entirely.

Example: r(HR_t, MAP_t) might be -0.15 (weak, same-hour).
         r(HR_t, MAP_t+2) might be -0.40 (strong, HR predicts MAP drop 2h later).
         Lag-0 Pearson completely misses the predictive relationship.

Fix: Compute cross-correlation at multiple lags using pandas .shift().
Plot the cross-correlogram for feature pairs of interest.
```

Failure Mode 4: Temporal lag blindness

✓ COMPLETE CODE: ALTERNATIVES TO PEARSON

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import spearmanr, pearsonr

VITALS = ['HR','O2Sat','SBP','MAP','Resp','Temp']

# ── 1. Spearman correlation matrix (robust to outliers + monotonic non-
linearity) ──
feat_df = df[VITALS].dropna()
spearman_mat = feat_df.rank().corr()  # Spearman = Pearson on ranks

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
import seaborn as sns

pearson_mat = feat_df.corr()
sns.heatmap(pearson_mat,  annot=True, fmt='.2f', cmap='coolwarm',
            center=0, ax=axes[0], vmin=-1, vmax=1)
axes[0].set_title('Pearson correlation (pooled, all rows)')

sns.heatmap(spearman_mat, annot=True, fmt='.2f', cmap='coolwarm',
            center=0, ax=axes[1], vmin=-1, vmax=1)
axes[1].set_title('Spearman correlation (pooled, all rows)')

plt.suptitle('Pearson vs Spearman: pooled over all patient-hours')
```

```
plt.tight_layout()
plt.savefig('correlation_comparison.png', dpi=120, bbox_inches='tight')
plt.show()

# ── 2. Within-patient correlation (controls for between-patient structure) ─
patient_corrs = []
for pid, grp in df.groupby('PatientID'):
    if len(grp) < 6:   # need at least 6 hours for a meaningful correlation
        continue
    sub = grp[VITALS].dropna()
    if len(sub) < 6:
        continue
    patient_corrs.append(sub.corr())

within_pt_corr = pd.concat(patient_corrs).groupby(level=0).mean()
print('Within-patient mean Pearson correlations:')
print(within_pt_corr.round(3))

# ── 3. Lagged cross-correlation for a specific feature pair ──────────────
def lagged_corr(series_x, series_y, max_lag=6):
    """Pearson r between x_t and y_{t+lag} for lag in [-max_lag,
+max_lag]."""
    results = {}
    for lag in range(-max_lag, max_lag + 1):
        shifted = series_y.shift(-lag)   # negative lag = y leads x
        valid   = series_x.notna() & shifted.notna()
        if valid.sum() > 10:
            r, _ = pearsonr(series_x[valid], shifted[valid])
            results[lag] = r
    return pd.Series(results)

# Example: HR at time t vs MAP at time t+lag
lc = lagged_corr(df['HR'], df['MAP'], max_lag=6)
plt.figure(figsize=(8, 4))
plt.bar(lc.index, lc.values, color='#5B2D8E', edgecolor='white')
plt.axhline(0, color='black', lw=0.8)
plt.xlabel('Lag (hours): HR leads MAP at positive lag')
plt.ylabel('Pearson r')
plt.title('Lagged cross-correlation: HR (t) vs MAP (t + lag)')
plt.tight_layout()
plt.savefig('lagged_corr.png', dpi=120, bbox_inches='tight')
plt.show()
```

✓ ALTERNATIVES COMPARISON TABLE

Method	What it measures	Advantages	Limitations	Best for
Pearson r	Linear association at lag 0	Fast, interpretable, standard	Assumes linearity, independence, no lags; outlier-sensitive	Normally distributed, independent samples — not ICU time series
Spearman r	Monotonic association at lag 0	Robust to outliers and non-linearity	Still lag-0 only; still ignores temporal structure	When you must use a correlation matrix and data has outliers

Within-patient Pearson	Within-patient linear association	Removes between-patient confounding	Still lag-0; computationally heavier	Studying within-patient physiology
Lagged cross-correlation	Association at multiple time offsets	Reveals temporal dependencies and lead-lag relationships	Requires stationarity; harder to summarise	Finding which features predict others N hours later
Granger causality	Whether X improves forecast of Y beyond Y's own history	Formal statistical test of temporal predictability	Assumes linearity; requires stationarity; over-sensitive in large N	Testing mechanistic temporal hypotheses
Distance correlation (dcor)	Any dependence including non-linear, non-monotonic	Detects arbitrary associations	Less interpretable; computationally heavy	When Spearman also misses the relationship

Key exam takeaway

The main answer to 'why is Pearson misleading?' is row independence — not non-linearity or outliers, which are secondary concerns. A student who gives only 'it assumes linearity' has missed the deeper issue. In longitudinal clinical data, the same patient contributes hundreds of correlated rows; Pearson treats these as 600 independent observations.

The practical recommendation for an EDA section in a clinical ML paper: report within-patient Spearman correlations averaged over patients, and show a lagged cross-correlogram for the 2-3 most clinically important feature pairs (HR vs MAP, Lactate vs Creatinine). These tell a richer story than a pooled Pearson matrix.

Q12 Site-bias analysis: Set A (BIDMC) vs Set B (Emory) — are there systematic differences?

HINT *The 2019 Challenge has two hospital sites. Plot the vital sign distributions separately for Set A and Set B. Are there systematic differences that could cause your model to be site-biased?*

✓ THE SHORT ANSWER

Yes, systematic differences between Set A and Set B are expected and typically found. The key diagnostic is to plot and statistically compare distributions of each vital sign separately by site, then test whether a model trained on one site generalises to the other. A site-biased model has learned hospital-specific patterns (equipment calibration, patient case mix, documentation practices) that do not reflect universal sepsis physiology and will perform poorly when deployed at a new institution.

✓ COMPLETE CODE: VISUALISATION + STATISTICAL TESTING

```
import matplotlib.pyplot as plt
import numpy as np
from scipy.stats import ks_2samp, mannwhitneyu
import pandas as pd

VITALS = ['HR','O2Sat','Temp','SBP','MAP','DBP','Resp']

# ── 1. Overlapping KDE plots for each vital by site ──────────────────────
fig, axes = plt.subplots(2, 4, figsize=(16, 8))
axes = axes.flatten()

for i, col in enumerate(VITALS):
    ax = axes[i]
    va = df_A[col].dropna()
    vb = df_B[col].dropna()

    ax.hist(va, bins=60, density=True, alpha=0.55,
            color='#2C5F8A', label='Set A (BIDMC)')
    ax.hist(vb, bins=60, density=True, alpha=0.55,
            color='#BF4F00', label='Set B (Emory)')

    ks_stat, p_val = ks_2samp(va, vb)
    sig = '***' if p_val < 0.05/len(VITALS) else ('*' if p_val < 0.05 else
'ns')
    ax.set_title(f'{col}  KS={ks_stat:.3f} {sig}', fontsize=9)
    ax.legend(fontsize=7)

axes[-1].axis('off')  # hide unused subplot
plt.suptitle('Vital sign distributions: Set A (BIDMC) vs Set B (Emory)',
fontsize=12)
plt.tight_layout()
plt.savefig('site_distributions.png', dpi=120, bbox_inches='tight')
plt.show()

# ── 2. Formal statistical comparison table ────────────────────────────────
bonferroni = 0.05 / len(VITALS)
rows = []
for col in VITALS:
```

```
    va = df_A[col].dropna().values
    vb = df_B[col].dropna().values
    ks_stat, ks_p   = ks_2samp(va, vb)
    mw_stat, mw_p   = mannwhitneyu(va, vb, alternative='two-sided')
    rows.append({
        'Feature': col,
        'Mean_A':  va.mean(),   'Median_A': np.median(va),
        'Mean_B':  vb.mean(),   'Median_B': np.median(vb),
        'KS_D':    ks_stat,     'KS_p':     ks_p,
        'MW_p':    mw_p,
        'Sig_Bonferroni': 'YES' if ks_p < bonferroni else 'no',
    })

stats_df = pd.DataFrame(rows)
print(stats_df.to_string(index=False, float_format='{:.4f}'.format))

# ── 3. Cross-site AUROC evaluation ──────────────────────────────────────
# Train on Set A only, test on Set B only
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing  import StandardScaler
from sklearn.metrics        import roc_auc_score

FEATURE_COLS = [c for c in df.columns
                if c not in ('PatientID','Hour','SepsisLabel','EarlyLabel')]

X_A = df_A[FEATURE_COLS].fillna(0).values
y_A = df_A['SepsisLabel'].values
X_B = df_B[FEATURE_COLS].fillna(0).values
y_B = df_B['SepsisLabel'].values

scaler   = StandardScaler()
X_A_sc   = scaler.fit_transform(X_A)
X_B_sc   = scaler.transform(X_B)  # same scaler -- simulates deployment

lr       = LogisticRegression(max_iter=1000, class_weight='balanced',
                              C=0.1, random_state=42)
lr.fit(X_A_sc, y_A)

# Evaluate on Set A (within-site) and Set B (cross-site)
probs_A  = lr.predict_proba(X_A_sc)[:,1]
probs_B  = lr.predict_proba(X_B_sc)[:,1]

auroc_A  = roc_auc_score(y_A, probs_A)
auroc_B  = roc_auc_score(y_B, probs_B)

print(f'Within-site AUROC (trained on A, tested on A): {auroc_A:.4f}')
print(f'Cross-site  AUROC (trained on A, tested on B): {auroc_B:.4f}')
print(f'Cross-site degradation: {auroc_A - auroc_B:.4f} AUROC points')
```

✓ EXPECTED FINDINGS AND INTERPRETATION

Feature	Typical KS stat	Likely significant (Bonferroni)?	Most probable clinical cause	Model impact if unaddressed
HR	0.05 – 0.15	Sometimes	Different patient acuity mix; Emory has higher trauma/surgical volume	Model over-alerts on Set B patients if Emory HR

				distribution is higher
O2Sat	0.03 – 0.08	Rarely	Equipment calibration; patient mix (COPD prevalence)	Small impact; O2Sat ranges are tightly constrained physiologically
Temp	0.03 – 0.12	Sometimes	Unit error (F vs C) would show huge D; case mix differences	If unit error: catastrophic. If case mix: moderate impact
SBP	0.04 – 0.10	Sometimes	Hypertensive patient prevalence; antihypertensive medication use patterns	Model may miss hypotension in patients with high-baseline BP at Emory
Resp	0.05 – 0.14	Often	COPD prevalence differences; documentation practice (continuous vs periodic charting)	Model may over-alert on Set B COPD patients if RR baseline is higher
Cross-site AUROC gap	3 – 8 AUROC points	N/A	Combined effect of all covariate shifts above	The overall impact — a gap > 5 points warrants investigating before deployment

✓ SOURCES OF SITE DIFFERENCES — A TAXONOMY

Source	What it produces in data	How to detect it
Patient case mix	Different distributions of age, comorbidities, admission diagnosis — BIDMC is a major teaching hospital with complex medicine; Emory has large cardiac and transplant programs	Compare patient demographics (Age, ICU unit type) between sites; compare comorbidity proxies (baseline Creatinine, Glucose)

Equipment calibration	Systematic offset in measured values — one hospital's blood pressure cuffs may read 3 mmHg higher	Look for constant mean differences between sites; KS test with large D and p~0 suggests calibration offset rather than distribution shape change
Documentation practices	Some ICUs chart vital signs every 30 min and average to hourly; others chart only when values change	Compare within-patient autocorrelation between sites; higher autocorrelation suggests less frequent real measurements
Lab assay methods	Different reagents, analysers, or reference ranges — Creatinine methods (Jaffe vs enzymatic) can differ by 0.2 mg/dL systematically	Look for systematic median differences in lab values; compare missingness rates (one site may order labs more frequently)
Sepsis definition adoption	One site may have adopted Sepsis-3 earlier; label prevalence differs	Compare sepsis rates between sites — a large difference (>3 percentage points) suggests definitional inconsistency

✓ MITIGATION STRATEGIES (IN ORDER OF INVASIVENESS)

Strategy 1 — Report cross-site AUROC as primary metric: Before any mitigation, report AUROC separately for (a) training on A, testing on A; (b) training on A, testing on B; (c) training on B, testing on A. These three numbers quantify the site-bias magnitude. A difference < 2 AUROC points is acceptable; > 5 AUROC points requires mitigation before deployment at a new site.

Strategy 2 — Include Site as a binary feature: Add a column Site = 0 (Set A) or 1 (Set B). This allows the model to learn site-specific offsets. The risk: the model may become even more site-dependent. This is appropriate only when you plan to deploy at both specific sites.

Strategy 3 — Site-specific standardisation (z-scoring within site): Z-score each feature using the site-specific mean and standard deviation before merging. This removes systematic calibration differences while preserving within-site variation. Implementation: scaler_A.fit(X_A); scaler_B.fit(X_B); then transform each site's data with its own scaler before combining. The downside: the site-specific scalers must be retained and applied at deployment time.

Strategy 4 — Importance weighting (domain adaptation): Estimate the ratio P(X | Site=A) / P(X | Site=B) using a density ratio estimator (e.g. a logistic regression trained to discriminate sites). Upweight examples from the source site to match the target distribution. This is the most principled approach for covariate shift but adds significant implementation complexity.

Key exam takeaway

The most important output of Q12 is not the distribution plots themselves but the cross-site AUROC gap. A model that achieves AUROC 0.87 within-site and 0.81 cross-site has a 6-point

generalisation gap. This is the number that matters when a hospital administrator asks 'will this model work at our institution?'

In any publication, always report: (1) which sites contributed to training data, (2) which sites were used for evaluation, and (3) whether the evaluation was within-site or cross-site. Most published sepsis prediction papers use within-site evaluation — making their reported AUROCs optimistically biased for real-world deployment.

Self-Assessment Checklist — Step 2

Check each item after completing Step 2. Items marked ★ are likely exam questions.

EDA Fundamentals

- ☐ I can explain why class balance must be measured at BOTH patient and hour level ★
- ☐ I know the 5 expected signal directions for vitals in sepsis (HR, Resp up; O2Sat, SBP, MAP down)
- ☐ I can produce a missing data heatmap and classify features into three missingness tiers
- ☐ I understand the difference between sensor dropout missingness and clinically informative missingness ★

Q9 — EtCO2 Missingness Strategy

- ☐ ★ I can defend the binary flag + imputation approach and explain why dropping is wrong
- ☐ I know what EtCO2 physically requires to be measured (intubation / ventilator)
- ☐ I can name 5 features beyond EtCO2 that deserve missingness flags and explain why for each
- ☐ I can write the two-line code to create a binary flag before imputation ★
- ☐ I know how to empirically validate whether a flag adds predictive value (AUROC comparison)

Q10 — Age-Stratified Trends

- ☐ ★ I can explain why elderly patients show blunted tachycardia (beta-blockers, autonomic dysfunction)
- ☐ I know which vital sign is most different between age groups and why
- ☐ I understand why the answer to 'do older patients deteriorate earlier?' is nuanced, not yes/no
- ☐ I can describe the model fairness concern and the correct mitigation (disaggregated AUROC) ★

Q11 — Pearson Correlation Limitations

- ☐ ★ I can name and explain all four failure modes of Pearson for ICU time series
- ☐ I know which failure mode is most important (row independence / temporal autocorrelation) ★
- ☐ I can describe Spearman correlation and when to use it over Pearson
- ☐ I can implement lagged cross-correlation using pandas .shift() and explain what the peak lag means ★
- ☐ I know what within-patient correlation is and why it differs from pooled correlation

Q12 — Site Bias Analysis

- ☐ ★ I can explain the four sources of site differences (case mix, calibration, documentation, assay)
- ☐ I can write the cross-site AUROC evaluation code (train on A, test on B) ★
- ☐ I know 4 mitigation strategies in order of invasiveness
- ☐ I understand why published within-site AUROCs are optimistically biased for real deployment ★
- ☐ I can identify a temperature unit error (F vs C) by looking at the distribution shift in the KS plot

How to use this cheatsheet responsibly

These answers represent thorough graduate-level responses, not minimal passing answers. For exams, aim to cover the core mechanism and one concrete example. For research papers or deployment proposals, use the full depth shown here. Items marked ★ should be mastered — they represent concepts that distinguish strong from weak answers.